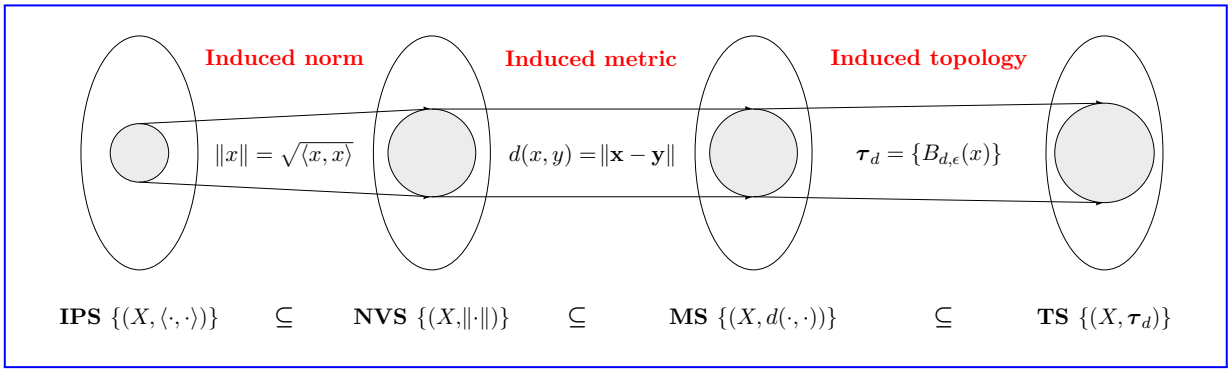

Figure 1: Induced Norm, Metric, and Topology

IPS	$--$	Inner Product Spaces	$(X, \langle \mathbf{x}, \mathbf{y}\rangle)$
NVS	$--$	Normed Vector Spaces	$(X, \|\mathbf{x}\|) = (X, \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle})$
MS	$--$	Metric Spaces	$(X, d(\mathbf{x}, \mathbf{y})) = (X, \|\mathbf{x} - \mathbf{y}\|) = (X, \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y}\rangle})$
TS	$--$	Topological Spaces	$(X, \boldsymbol{\tau}_d) = (X, \{B_{d,\epsilon}(\mathbf{u})\})$, where $U \subseteq X$ and $(\forall \mathbf{u} \in U,\ \exists \epsilon > 0) : B_{d,\epsilon}(\mathbf{u}) \subseteq U$

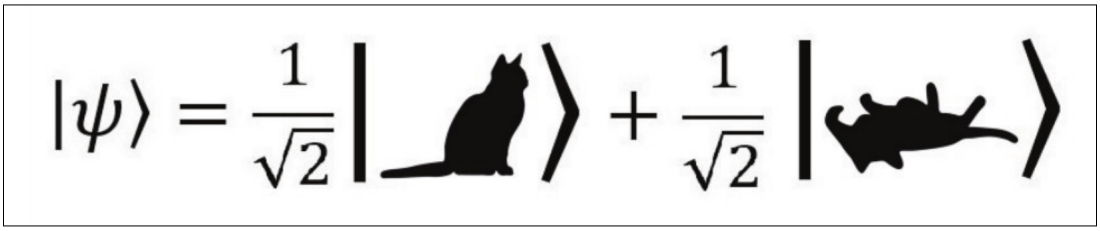

Figure 2: Quantum superposition and Schrödinger's cat

$n=\langle\ ,\ \rangle$
$G(\ )$
$\ \ \ \

0.6249999121401877

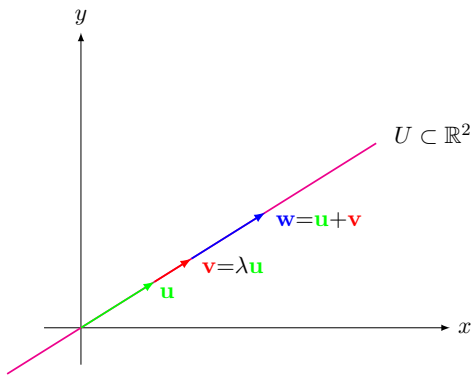

Figure 5: The line U is a linear subspace of $\mathbb{R}^2$

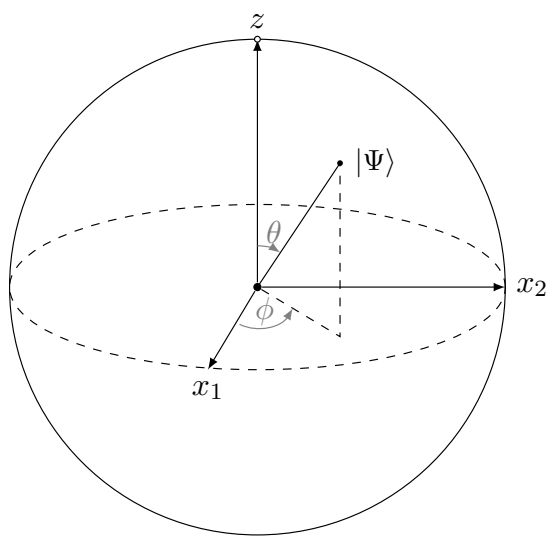

Figure 6: The Bloch Sphere

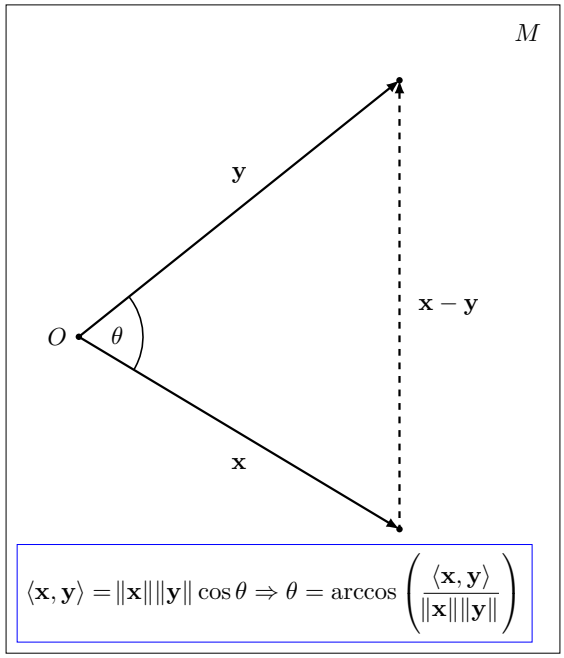

Figure 7: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

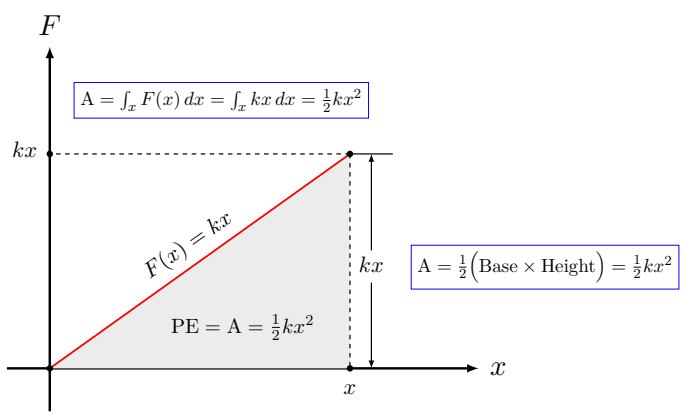

Figure 8: Potential Energy of a Simple Harmonic Oscillator

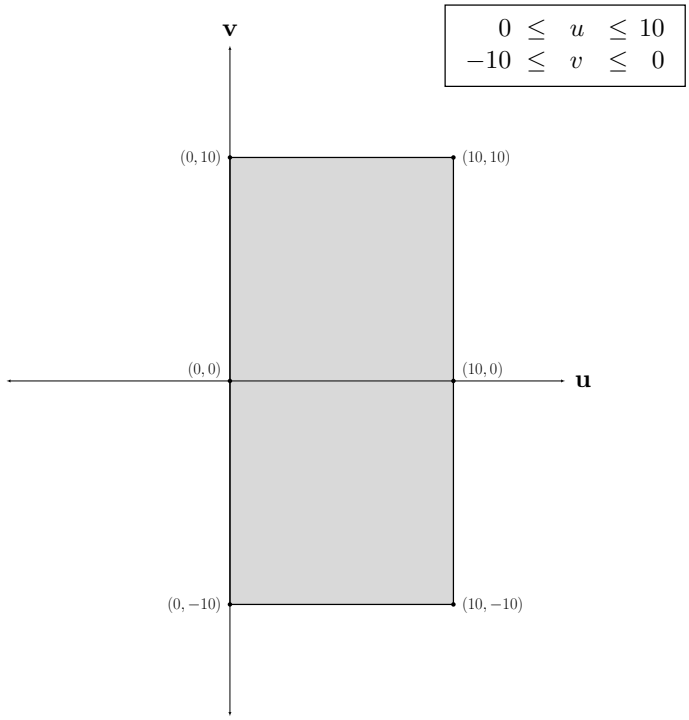

Figure 9: The transformed parallelogram is a rectangle in the uv-plane

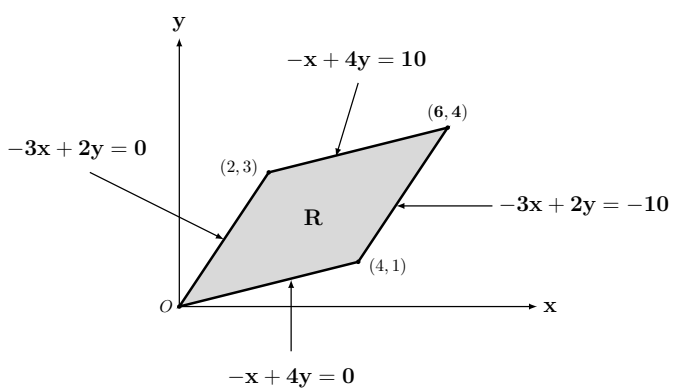

Figure 10: Equations of the Sides of the Parallelogram

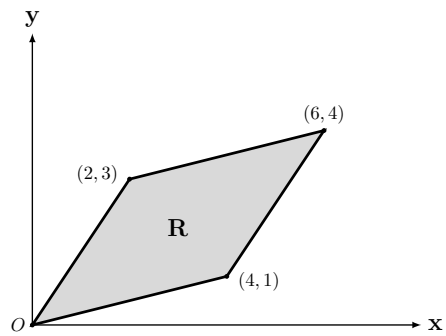

Figure 11: Parallelogram in the xy-plane

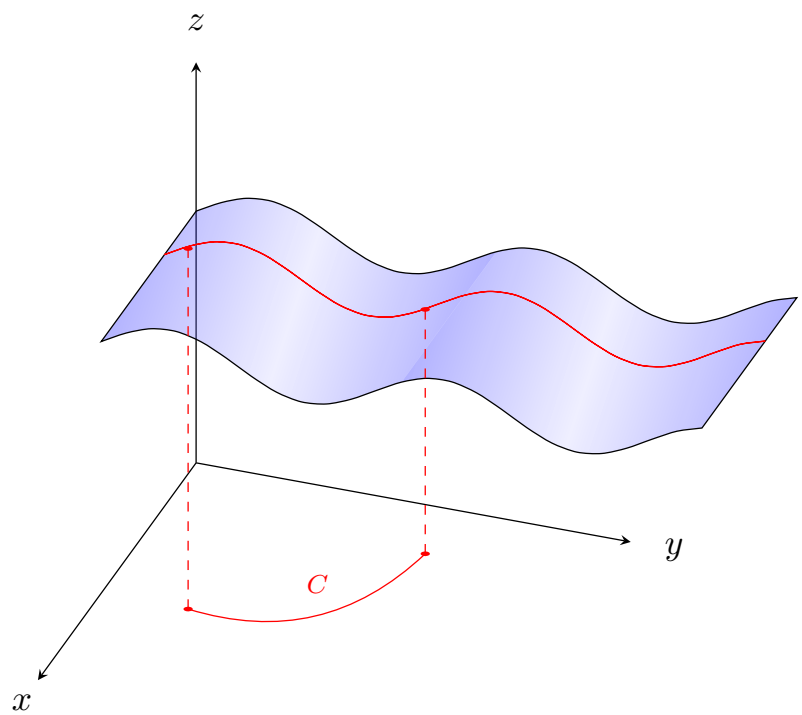

Figure 12: 3D

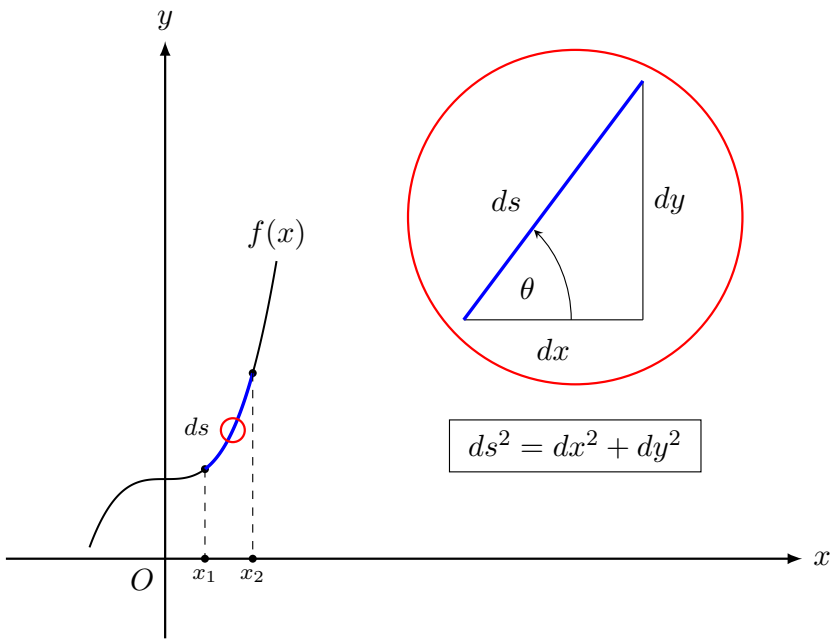

Figure 13: $f(x)$, ds and the Pythagorean Theorem

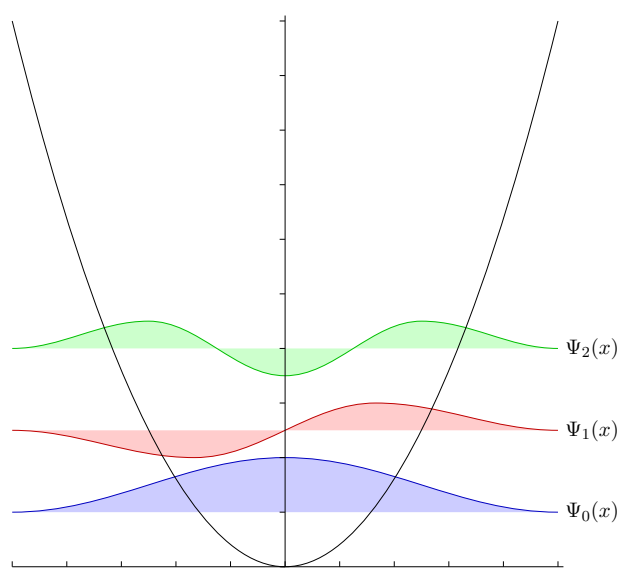

Figure 14: Wave Functions

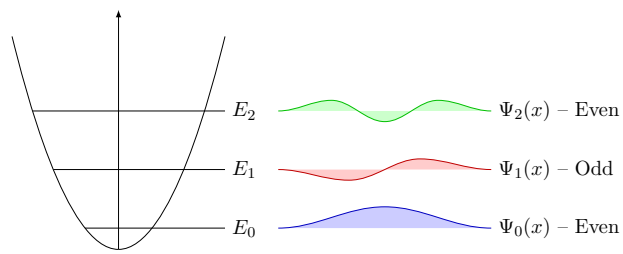

Figure 15: Wave Functions

9^2	$=$	81	$\longrightarrow$	$8+1$	$=$	9
45^2	$=$	2025	$\longrightarrow$	$20+25$	$=$	45
703^2	$=$	494209	$\longrightarrow$	$494+209$	$=$	703
7777^2	$=$	60481729	$\longrightarrow$	$6048+1729$	$=$	7777
857143^2	$=$	734694122449	$\longrightarrow$	$734694+122449$	$=$	857143

Figure 16: A Few Example Kaprekar Numbers

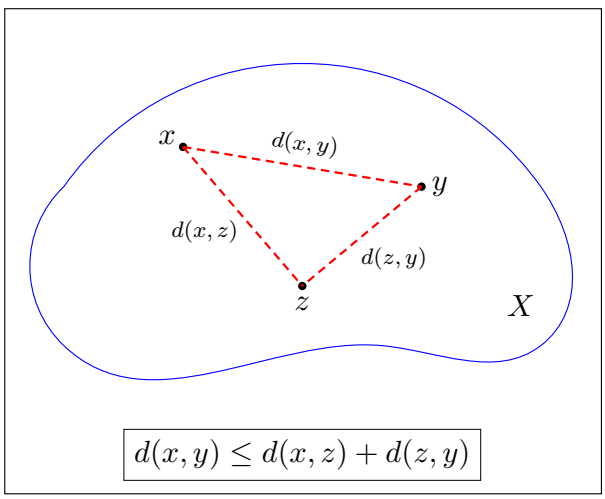

Figure 17: The Triangle Inequality

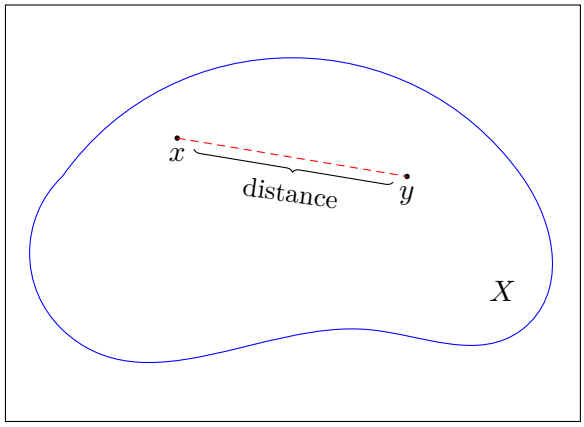

Figure 18: X, x and y

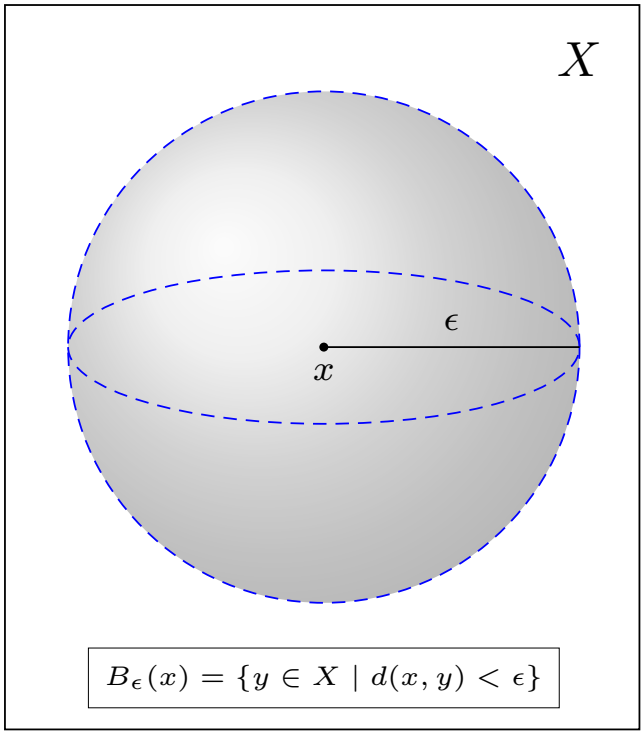

Figure 19: $B_\epsilon(x)$ is an open epsilon ball centered at x with radius ϵ

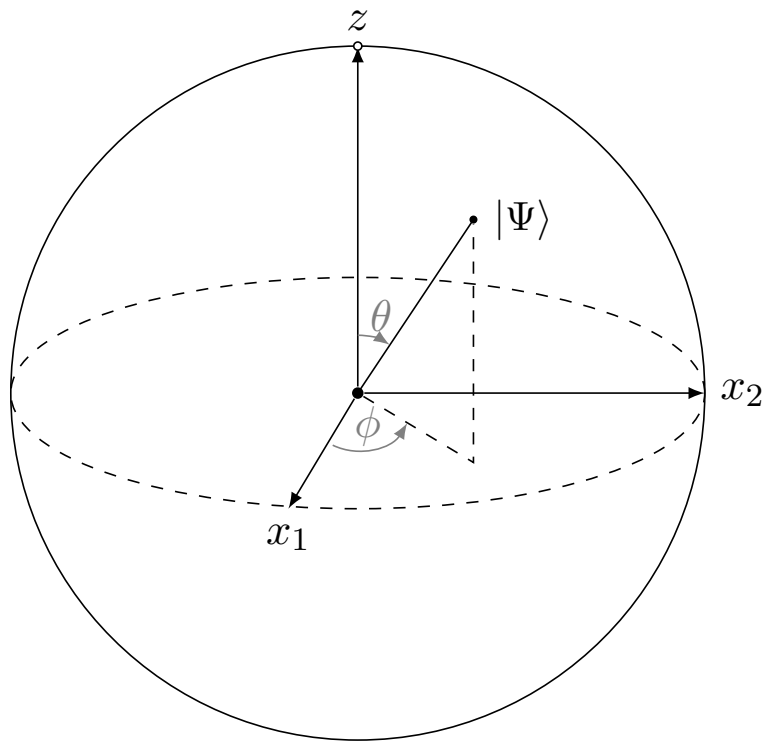

Figure 20: The Bloch Sphere

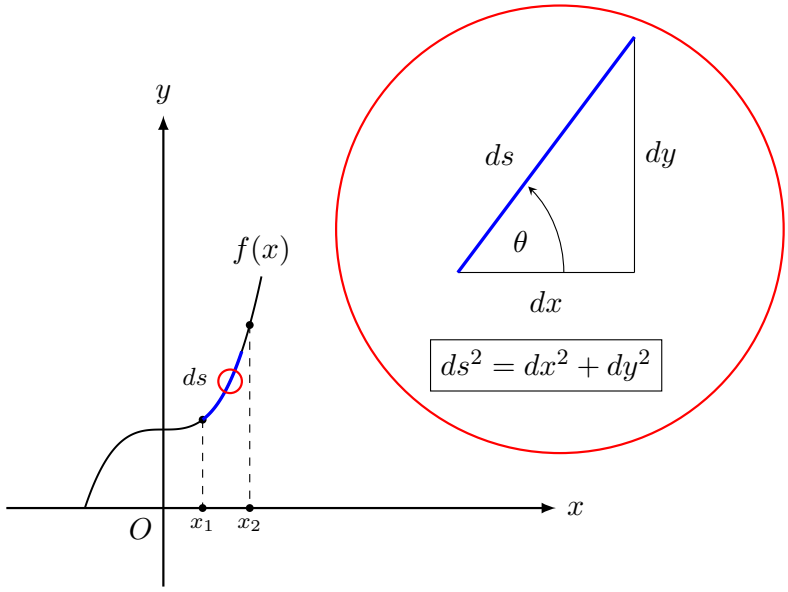

Figure 21: $f(x)$, ds and the Pythagorean Theorem

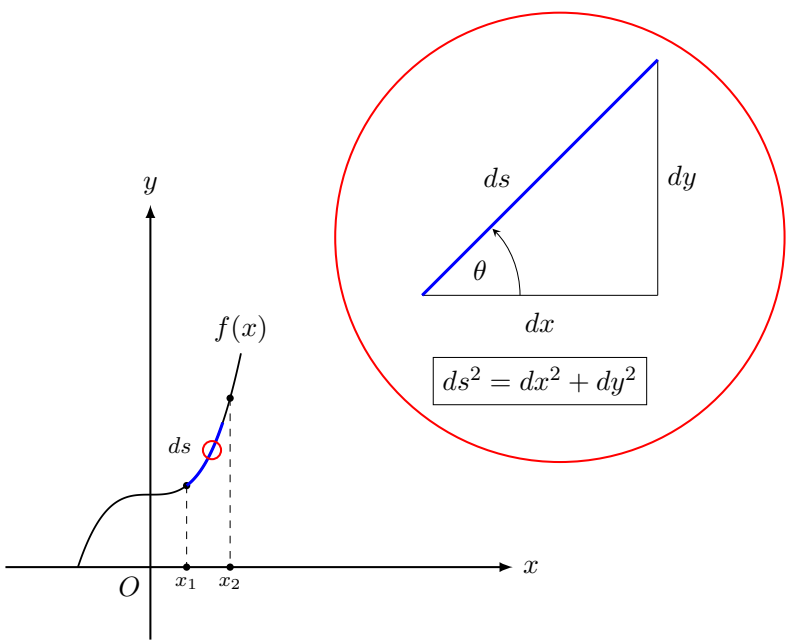

Figure 22: $f(x)$, ds and the Pythagorean Theorem